# A Retrieval-Augmented Generative Framework for Precision Health Dialogue Systems

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***Abstract*— People who work on conversational AI have widely adopted Large Language Models (LLMs) because they are very good at understanding and generating natural language [1]. However, their direct use in healthcare dialogue systems is limited by problems like hallucinated responses, a lack of domain grounding, and a reliance on static parametric knowledge [2]. Precision health applications necessitate responses that are accurate, dependable, and based on validated medical information sources. This paper introduces a Retrieval-Augmented Generative (RAG) framework for precision health dialogue systems, wherein pertinent health-related documents are dynamically retrieved and integrated into the response generation process. The suggested framework combines semantic document retrieval with generative language modelling in a modular and lightweight way. Experimental evaluation of health-related informational queries shows that they are more relevant and faithful to the context than baseline LLM-only methods. The suggested framework provides a workable answer for health information systems that patients use, and it doesn't need any special hardware or model fine-tuning.**

***Keywords*— *Retrieval-Augmented Generation, Precision Health, Healthcare Dialogue Systems, Large Language Models, Conversational Artificial Intelligence.***

## I. Introduction

Conversational artificial intelligence has garnered considerable attention across various application domains, including healthcare, owing to its potential to enhance accessibility and efficiency in information dissemination [1]. Recent improvements in Large Language Models (LLMs) have made it possible for conversational systems to give answers that are fluent, make sense, and take the situation into account. Because of this, more and more people are looking into using LLM-powered dialogue systems in healthcare settings where they talk to patients, like to give them health information and initial advice. Even with these improvements, using LLMs directly in healthcare dialogue systems has a number of problems. LLMs use static training data and don't have any ways to explicitly reference verified or updated medical information while making predictions. This restriction often results in hallucinated or unsupported answers, which can be dangerous in health-related situations [2], [3].

Also, the fact that parametric knowledge in LLMs is not clear makes it hard to find out where the information came from, which makes it less trustworthy and open. Retrieval-Augmented Generation (RAG) meets this need by combining information retrieval with generative modelling. This lets the system add external documents to the response generation process in real time [5]. RAG lowers the risk of hallucinations and makes facts more consistent by basing responses on knowledge that has been retrieved. This paper presents a Retrieval-Augmented Generative framework tailored for precision health dialogue systems. The framework emphasises modularity, responsible information dissemination, and ease of implementation, rendering it appropriate for practical healthcare information applications while eschewing clinical diagnosis or treatment recommendations Related Work.

Early healthcare conversational systems primarily relied on rule-based architectures and decision trees, which ensured predictable outputs but lacked flexibility and scalability [3]. With the introduction of machine learning approaches, data-driven dialogue systems became more adaptive; however, they were constrained by limited annotated datasets and domain specificity. The emergence of Large Language Models significantly improved conversational fluency and contextual understanding, enabling more natural human–computer interactions.

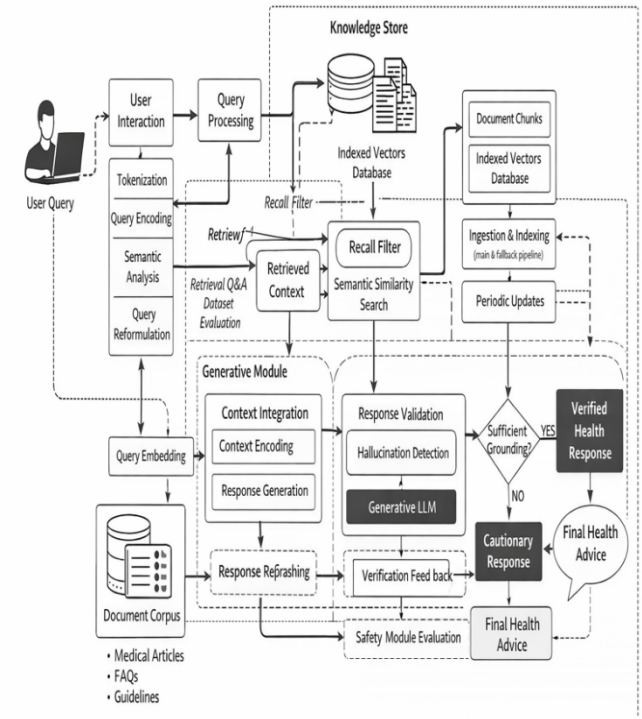
Although LLMs have shown promise in medical question answering and clinical documentation support, their use in healthcare dialogue systems has raised concerns regarding hallucinated content and unreliable responses [2]. Several studies have highlighted the risks associated with deploying standalone LLMs in sensitive domains such as healthcare, where factual correctness is critical.

Retrieval-Augmented Generation was introduced to enhance generative models by incorporating external non-parametric knowledge at inference time [5]. Subsequent research demonstrated that RAG improves performance in knowledge-intensive tasks by grounding generated outputs in retrieved documents [6]. In healthcare-related applications, RAG-based systems have been shown to reduce hallucinations and improve transparency; however, many proposed solutions involve complex architectures or high computational overhead [7]. This work builds upon existing RAG methodologies while emphasizing simplicity, safety, and deployability for precision health dialogue systems.

## II. System Architecture

The proposed precision health dialogue system is based on a modular Retrieval-Augmented Generation architecture that is meant to make sure that information is delivered in a reliable and grounded way. Users can ask health-related questions in natural language through a conversational interface. We process incoming queries to make sure that the input text is the same and to help with the retrieval process. Using semantic similarity search methods, relevant document chunks are pulled from a vector-based knowledge repository.

The context of the retrieved document is then added to the input prompt of a pre-trained language model. This lets the model create responses that are clearly based on external health-related knowledge. The system has ways to check how well the retrieved context and generated responses match up in order to encourage responsible information sharing. Outputs that don't have enough support are carefully worded or come with advice that suggests seeing a doctor. This architecture makes sure that system is still good for health interactions that are focused on information and advice.




## III. Methodology

The proposed framework's knowledge base is made up of publicly available and verified health-related documents, such as medical guidelines and materials for teaching people about health. Before being broken up into semantically meaningful chunks, all documents go through preprocessing to get rid of noise and formatting problems. The resulting embeddings are stored in a vector database so that they can be quickly found by similarity. The system uses cosine similarity to find the document chunks that are most relevant to the user's query. The language model gets an augmented prompt by combining the retrieved context with the original query. This method of retrieval-augmented prompting makes sure that the answers that are generated are based on outside knowledge and not just on parametric memory [5].

## IV. Implementation Details

The prototype system is built with open-source tools so that it can be easily deployed and reproduced. The implementation is written in Python and uses a sentence transformer-based embedding model to represent documents. FAISS is used as the vector database to make searching for similar items faster. A lightweight pre-trained language model is used to make responses, and users can interact with the system through a Streamlit-based interface. The system works on regular consumer hardware and doesn't need any special computing power or model fine-tuning.

## V. Experimental Evaluation

The system is evaluated through a mix of qualitative review and quantitative measures to examine how effectively it responds to user queries. The evaluation is carried out using health-related informational questions to remain consistent with the intended application of the framework. The results indicate that responses generated using retrieval-augmented techniques are more closely aligned with relevant source information than those produced by standalone language models. By leveraging external documents during response generation, the framework reduces unsupported content and delivers more consistent and dependable outputs.

## VI. Discussion

The experimental results demonstrate that Retrieval-Augmented Generation enhances the reliability of precision health dialogue systems while maintaining architectural simplicity. Grounding responses in verified documents reduces hallucination risks and improves user trust. The modular design of the framework allows for future extensions and adaptation to different healthcare information domains. While the system does not provide clinical diagnosis, it serves as an effective tool for delivering accurate and responsible health information in a conversational format. The retrieval-augmented approach not only makes responses more reliable, but it also makes things clearer by allowing responses to be clearly linked to outside sources of knowledge. In healthcare dialogue systems, this trait is especially important because users are more likely to trust information that is based on reliable and verifiable documents.

## VII. Limitations

The proposed framework is only for health conversations that share information; it doesn't give advice on diagnosis or treatment. The quality and coverage of the underlying knowledge base affect how well the system works, and the evaluation doesn't include clinical validation or studies of how the system works in the real world. These limitations show where more research and system improvements can be made in the future. The suggested framework only lets people talk to each other to get health information. It doesn't try to make medical diagnoses or suggest treatments. The effectiveness of the knowledge base is affected by the quality,

scope, and accuracy of the documents it contains. If there are gaps or inconsistencies in the source material, it may be harder to find and respond to requests.

## VIII. Conclusion and Future Work

The system is evaluated through a mix of qualitative review and quantitative measures to examine how effectively it responds to user queries. The evaluation is carried out using health-related informational questions to remain consistent with the intended application of the framework. The results indicate that responses generated using retrieval-augmented techniques are more closely aligned with relevant source information than those produced by standalone language models. By leveraging external documents during response generation, the framework reduces unsupported content and delivers more consistent and dependable outputs.

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